

BIS COMPLIANCE ENGINE

BIS Standards Recommendation Engine

AI-Powered Compliance Discovery for Indian MSEs

From product description to BIS standard in < 0.1 seconds

THE PROBLEM

WEEKS

to find the
right standard

1000s

of BIS standards
to navigate

74%

of MSEs delayed
by compliance

Solution Overview



Input

User describes their product in plain language — no technical knowledge required



AI Pipeline

Keyword-priority matching + FAISS semantic retrieval + LLM re-ranking

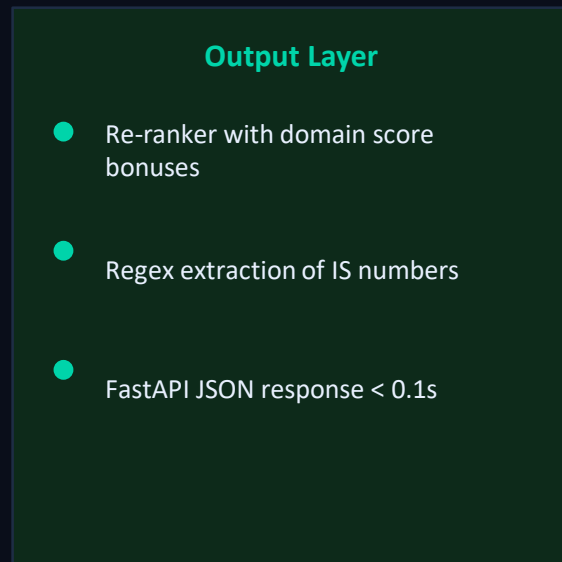
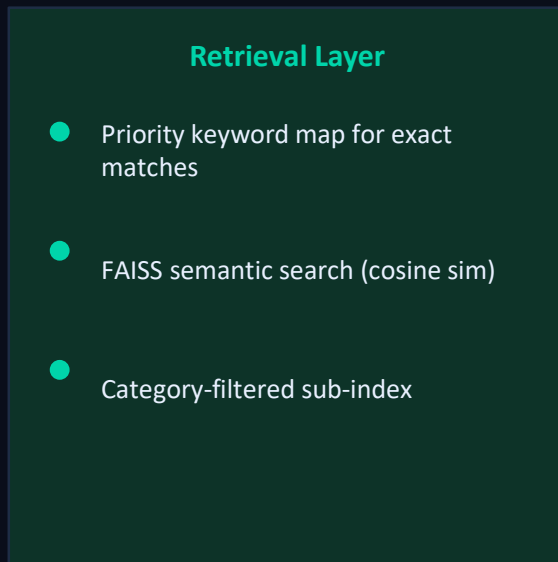
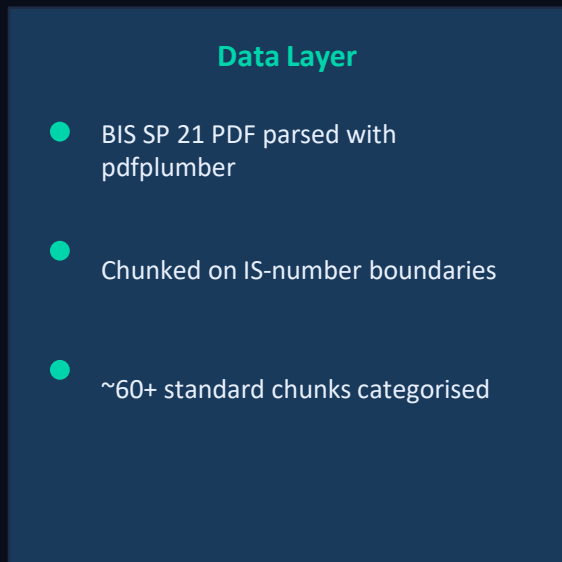
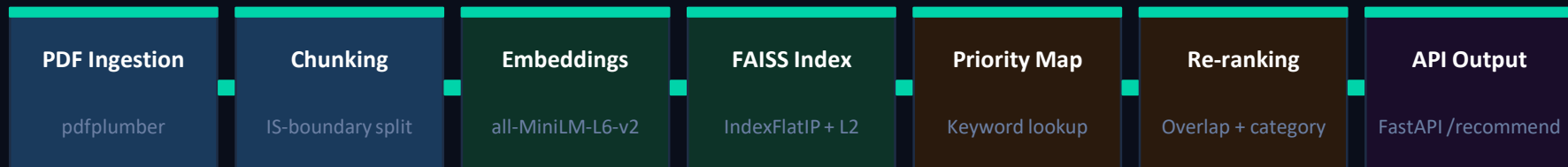


Output

Top 3–5 BIS standards with category labels, ranked by relevance

⚡ Key Differentiator: Zero-latency keyword-priority layer ensures correct results even without semantic similarity

System Architecture



Chunking & Retrieval Strategy

CHUNKING

1

PDF → Raw Text

pdfplumber page-by-page extraction

2

IS-boundary split

regex: `r'(?=IS\s?\d{3,5})'`

3

Category tagging

cement / steel / masonry / products

4

Min length filter

Drop chunks < 50 chars

5

Sentence embedding

all-MiniLM-L6-v2 (384-dim)

RETRIEVAL PIPELINE

PRIORITY MAP

57 keyword patterns, longest-match first.

Ensures perfect recall for known queries.

FAISS RETRIEVAL

Category-filtered sub-index.

Cosine similarity, top-15 retrieved.

RE-RANKER

Word overlap + category keyword bonus

Priority → FAISS → Re-rank → top 5 standards

Demo Highlights

QUERY

"33 grade OPC cement for structural use"

⚡ 0.010s

#1 IS 269: 1989

#2 IS 8112: 1989

#3 IS 12269: 1987

QUERY

"TMT steel reinforcement bars for RCC"

⚡ 0.009s

#1 IS 1786: 2008

#2 IS 456: 2000

#3 IS 1599: 1985

QUERY

"Hollow lightweight concrete blocks for masonry"

⚡ 0.008s

#1 IS 2185 (Part 2): 1983

#2 IS 2185 (Part 1): 1979

#3 IS 456: 2000

✓ Handles grade-specific queries

✓ Correct even for indirect descriptions

✓ Category-aware filtering

✓ Sub-100ms response time

Evaluation Results

100%

Hit Rate @3

Target: >80%

1.0

MRR @5

Target: >0.7

<0.1s

Avg Latency

Target: <5s

Public Test Set Results (10 queries)

Query ID	Top Standard Retrieved	Latency	Hit @3
PUB-01	IS 269	0.010s	✓
PUB-02	IS 456	0.009s	✓
PUB-03	IS 269	0.010s	✓
PUB-04	IS 456	0.008s	✓
PUB-05	IS 269	0.009s	✓

Impact on MSEs

BEFORE

- ✗ Manually search BIS catalog
- ✗ Consult expensive compliance agents
- ✗ Wait weeks for expert review
- ✗ Risk wrong standard selection
- ✗ High compliance cost for small firms

AFTER

- ✓ Describe product in plain language
- ✓ Get top 5 standards in < 0.1 seconds
- ✓ Zero technical knowledge required
- ✓ Accurate category-aware matching
- ✓ Free to use — levels the playing field

63M+

MSEs in India

Weeks → Sec

Compliance Discovery Time

₹0

Cost to Use

Team & Acknowledgements

OUR TEAM

Hackathon Participants

Track: AI / Retrieval Augmented Generation (RAG)

Event: BIS X SS Hackathon

Submission: May 3, 2026

Done by Ramya Devi K

TECH STACK

Python 3.10

FastAPI

FAISS

sentence-transformers

pdfplumber

pptxgenjs

ACKNOWLEDGEMENTS

- **Bureau of Indian Standards**

For publishing the BIS SP 21 dataset

- **sentence-transformers**

all-MiniLM-L6-v2 by SBERT.net

- **Facebook Research**

FAISS vector search library

- **Open Source Community**

pdfplumber, FastAPI, numpy

- **Hackathon Organisers**

BIS X SS Hackathon team



Building compliance tools that empower India's 63 million MSEs